

BER Performance Comparison of Classical and AI-Based BPSK Detection over Rayleigh Fading Channels

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Abstract

Reliable signal detection is a fundamental problem in wireless communication systems because fading and noise can cause transmitted bits to be detected incorrectly. This project studies the bit error rate (BER) performance of Binary Phase Shift Keying (BPSK) transmission over a Rayleigh fading channel. First, a classical coherent detector with perfect channel state information (CSI) is implemented as the baseline. Its BER performance is analyzed theoretically and validated through Python-based Monte Carlo simulation. Next, a simple artificial intelligence (AI)-based detector is introduced using a feedforward neural network trained on features extracted from the equalized received signal. The BER performance of the classical detector and the neural network detector is then compared over a range of signal-to-noise ratio (SNR) values. The results show that the simulation closely matches the theoretical BER for coherent BPSK over Rayleigh fading, confirming the correctness of the baseline model. The AI-based detector achieves BER values very close to the classical detector, but it does not provide a significant improvement under the ideal assumptions used in this project. This suggests that, for the considered setting with perfect CSI, the classical detector remains highly effective and the added complexity of a neural network may not be strongly justified.

Keywords— BPSK, Rayleigh fading, BER, coherent detection, neural network detector, machine learning, wireless communications

1. Introduction

Wireless communication systems operate in environments where transmitted signals are affected by noise, attenuation, and multipath fading. Unlike wired channels, wireless channels vary over time and space, which makes reliable detection of transmitted information more challenging. One of the most common ways to evaluate the reliability of a digital communication system is through the bit error rate (BER), which measures the probability that a transmitted bit is detected incorrectly.

Among digital modulation methods, Binary Phase Shift Keying (BPSK) is one of the simplest and most widely studied schemes. Because of its simplicity, BPSK is frequently used in theoretical analysis and simulation of communication systems. In fading channels, especially Rayleigh fading channels, the received signal amplitude changes randomly due to multipath propagation. As a result, even if the average SNR is fixed, the instantaneous channel quality varies and causes detection errors.

This project focuses on the performance of BPSK over a Rayleigh fading channel. The first goal is to build a correct classical baseline using coherent detection with perfect channel state information. The second goal is to validate the baseline by comparing theoretical BER and simulated BER using Monte Carlo methods. The third goal is to introduce a simple AI-based detector and compare its BER performance with the classical detector under the same communication conditions.

The motivation of this work is to understand whether a simple neural network can provide meaningful performance improvement over an already strong classical detector. This question is important because machine learning methods are increasingly explored in communication systems, but their usefulness depends on whether the extra complexity produces practical benefits.

2. Problem Definition and Motivation

The problem studied in this project is the detection of BPSK symbols transmitted over a Rayleigh fading wireless channel in the presence of additive white Gaussian noise (AWGN). The receiver must determine whether the transmitted bit was 0 or 1 based on the noisy received signal. The performance metric used is BER.

This problem is important because wireless systems commonly experience fading, and reliable symbol detection is essential for maintaining communication quality. In a fading environment, the received signal can be severely weakened, which increases the likelihood of detection errors. Therefore, understanding BER behavior under fading is a basic and important step in wireless system design.

The project uses a classical coherent detector as the baseline because it represents a standard analytical solution. Then, a simple neural network detector is trained using features from the received signal. The purpose is not to build a highly advanced deep learning model, but to evaluate whether a small AI model can match or improve the performance of the classical detector under fair comparison conditions.

3. Theoretical Background

3.1 BPSK Modulation

Binary Phase Shift Keying is a binary digital modulation technique in which each bit is mapped to one of two possible signal values. In this project, the mapping is:

$$\mathbf{b} \in \{0,1\}$$

$$\mathbf{x} = 2\mathbf{b} - 1$$

Thus,

$$0 \rightarrow -1 \quad \text{and} \quad 1 \rightarrow +1.$$

This representation allows simple threshold-based detection at the receiver.

3.2 Rayleigh Fading Channel Model

In a Rayleigh fading channel, the channel coefficient is modeled as a complex Gaussian random variable:

$$\mathbf{h} \sim \text{CN}(\mathbf{0}, 1)$$

The received signal is given by:

$$\mathbf{y} = \mathbf{h}\mathbf{x} + \mathbf{n}$$

where $\mathbf{x}$ is the transmitted BPSK symbol, $\mathbf{h}$ is the Rayleigh fading channel coefficient, and $\mathbf{n}$ is complex AWGN.

The noise term is modeled as

$$\mathbf{n} \sim \text{CN}(\mathbf{0}, \sigma^2)$$

where the noise variance depends on the SNR value.

3.3 Coherent Detection with Perfect CSI

If the receiver knows the channel coefficient $\mathbf{h}$, it can equalize the received signal as:

$$\mathbf{y}_{\text{eq}} = \mathbf{y} / \mathbf{h}$$

Substituting the received signal model gives:

$$\mathbf{y}_{\text{eq}} = \mathbf{x} + \mathbf{n} / \mathbf{h}$$

The decision rule for coherent BPSK detection is: if $\text{Re}(\mathbf{y}_{\text{eq}}) > 0$, detect bit 1; otherwise, detect bit 0.

This detector is simple and effective under perfect CSI.

3.4 Theoretical BER

For coherent BPSK over a Rayleigh fading channel, the theoretical BER is:

$$\mathbf{P}_b = 0.5 \times (1 - \sqrt{\gamma / (1 + \gamma)})$$

where γ is the average SNR in linear scale. This expression serves as the analytical benchmark for simulation validation.

4. System Model

The communication system used in this project consists of the following stages:

- Random bit generation
- BPSK modulation
- Rayleigh fading channel
- AWGN addition
- Equalization using perfect CSI
- Classical coherent detection
- AI-based detection using neural network classification

The assumptions are perfect channel state information at the receiver, Rayleigh flat fading, complex AWGN, BPSK modulation, and BER as the performance metric.

5. Python Simulation of the Classical Baseline

The baseline simulation was implemented in Python using NumPy and Matplotlib. For each SNR value, 100000 random bits were generated, mapped to BPSK symbols, passed through a Rayleigh fading channel, corrupted by complex AWGN, equalized using the channel coefficient, and finally detected using the sign of the real part of the equalized signal.

This process is a Monte Carlo simulation because many random realizations of bits, channels, and noise are used to estimate average system performance.

5.1 Simulation Parameters

Parameter	Value
Number of bits per SNR point	100000
SNR range	0 dB to 20 dB
SNR step size	2 dB
Modulation	BPSK
Channel model	Rayleigh flat fading
Noise model	Complex AWGN
Detection type	Coherent detection with perfect CSI

Performance metric	BER
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5.2 Baseline Validation

The first important result of the project is the comparison between theoretical BER and simulated BER. The generated figure shows that the simulation closely follows the theoretical curve over the entire tested SNR range. This confirms that the classical baseline was implemented correctly.

As expected, BER decreases as SNR increases. At low SNR, the noise and fading cause more detection errors. At high SNR, the received signal becomes more distinguishable and the BER becomes much smaller.

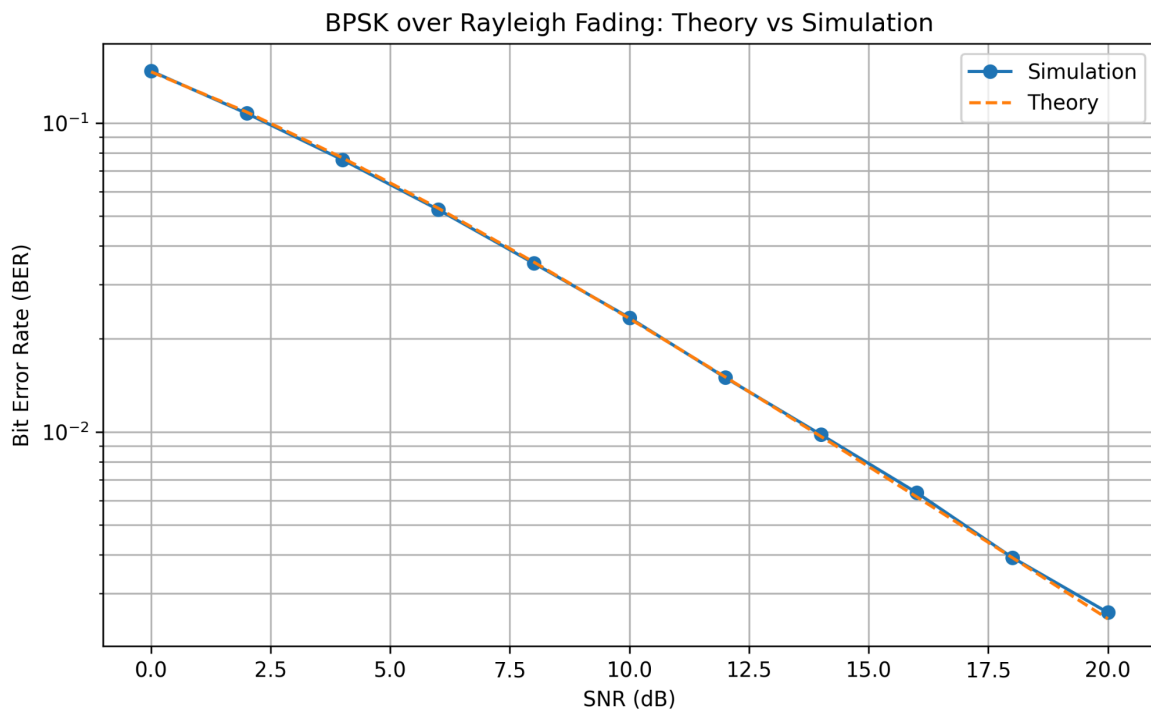


Figure 1. BER performance of coherent BPSK over Rayleigh fading: theoretical vs simulated results.

6. AI-Based Enhancement

After validating the classical baseline, a simple AI-based detector was introduced. The goal was to compare a machine learning detector with the classical coherent detector under the same channel conditions.

6.1 Motivation for the AI Detector

The classical detector uses a known analytical decision rule. In contrast, the AI detector attempts to learn the mapping from received signal features to transmitted bits directly from data. This can be useful when the channel model is difficult to derive, imperfectly known, or mismatched. In this project, the AI detector is intentionally kept simple.

6.2 Input Features and Labels

The feature vector for the neural network is formed from the equalized received signal:

$$\mathbf{X} = [\text{Re}(\mathbf{y}_{\text{eq}}), \text{Im}(\mathbf{y}_{\text{eq}})]$$

The labels are the original transmitted bits, where $b \in \{0,1\}$.

6.3 Model Structure

A simple feedforward neural network was implemented using MLPClassifier from scikit-learn. The model contains two hidden layers with 32 and 16 neurons, respectively, and uses the ReLU activation function.

6.4 Training and Testing Setup

For each SNR value, a dataset was generated from the equalized received signal, then split into training and testing sets using a 70/30 split. The neural network was trained on the training set, evaluated on the test set, and BER was computed from the prediction errors.

6.5 Performance Metric

The same BER metric was used for both the classical and AI-based detectors to ensure a fair comparison.

$$\text{BER} = \text{number of incorrectly detected bits} / \text{total number of bits}$$

7. Results and Discussion

7.1 Theory vs Simulation

The baseline result showed that the simulated BER matches the theoretical BER very closely across the SNR range from 0 dB to 20 dB. This validates the correctness of the classical simulation and establishes a reliable baseline for comparison with the AI detector.

7.2 Classical vs AI-Based Detection

The second result compares the classical coherent detector and the neural network detector. The BER values obtained from the experiment are summarized below.

SNR (dB)	Classical BER	AI BER
0	0.147300	0.149800
2	0.107390	0.104967
4	0.076030	0.075600
6	0.052440	0.052400
8	0.035140	0.035633
10	0.023410	0.023133
12	0.014990	0.014900
14	0.009800	0.009567
16	0.006360	0.006500
18	0.003920	0.003900
20	0.002600	0.002533

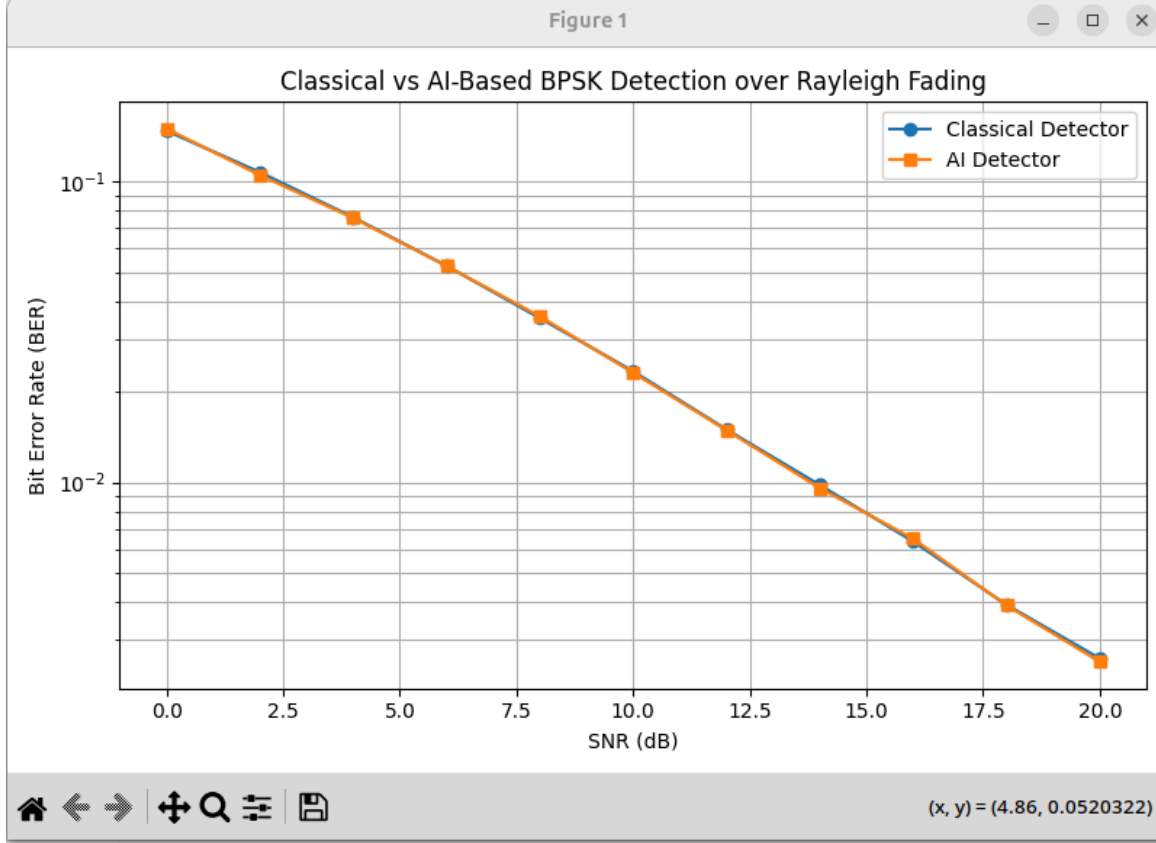


Figure 2. BER comparison of classical coherent detection and AI-based neural network detection.

7.3 Why the AI Detector Does Not Strongly Outperform

This result is reasonable because the classical detector is already very strong in this setting. The equalized received signal contains the essential information needed for decision making, and the classical rule based on the sign of the real part is already close to optimal for coherent BPSK detection with perfect CSI. Therefore, the neural network mostly learns a decision boundary similar to the classical threshold rule.

7.4 Complexity Discussion

The classical detector is mathematically simple, requires no training data, and has very low computational complexity during detection. In contrast, the AI-based detector introduces extra steps such as dataset preparation, train/test splitting, model training, and parameter tuning. Since

the BER performance is nearly the same in this project, the added AI complexity is not strongly justified under ideal conditions with perfect CSI.

7.5 Practical Insight and Limitations

Although the AI detector does not provide major improvement here, this does not mean AI has no role in communication systems. AI may become more useful under more difficult conditions such as imperfect or noisy channel estimation, model mismatch, non-Gaussian noise, or more complex modulation. The main limitations of this project are the assumption of perfect CSI, the use of simple features, the intentionally small neural network, training performed independently at each SNR point, and the use of BER as the main comparison metric.

8. Conclusion

This project investigated the BER performance of BPSK over a Rayleigh fading channel using both a classical coherent detector and a simple AI-based detector. The classical baseline was first implemented and validated by comparing Monte Carlo simulation results with the theoretical BER expression. The close match between theory and simulation confirmed that the baseline model was correct.

A feedforward neural network detector was then introduced using the real and imaginary parts of the equalized received signal as input features. The BER comparison showed that the AI-based detector achieved performance very close to the classical detector across the tested SNR range.

The main conclusion is that, in the considered ideal setting with perfect CSI, the classical coherent detector is already highly effective, and a simple neural network does not provide a significant BER improvement. Therefore, the extra complexity of the AI-based method is not strongly justified in this specific scenario.

However, the project also provides a useful practical insight: AI-based detection may become more valuable when the communication environment becomes more realistic and difficult, such as under-channel estimation errors, interference, or model uncertainty. This makes the current work a good starting point for future extensions.

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